

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2021

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2021 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. A irrigation kit is marked at \$1000 in Masvingo. <<
>>(a) 20% discount. Discounted price. [3]<<
>>(b) Then add VAT 15%. Amount paid. [3]<<
>>(c) Deposit 40% of the final amount. Balance remaining. [3] **[9]**

2. A rectangular garden in Mucheke measures 18 m by 8 m. A path of width 2 m runs all around the outside. <<
>>(a) Area of the garden. [2]<<
>>(b) Area of the path. [3]<<
>>(c) A circular bed of radius 7 m is later dug. Area of the bed, $\pi = 22/7$. [3] **[8]**

3. $AB \parallel CD$ with transversal EF in a street plan of Masvingo. <<
>>(a) Angle APQ = 65° . State the corresponding angle at Q, with a reason. [2]<<
>>(b) Find the co-interior angle to 65° . [2]<<
>>(c) A triangle has angles 65° , 54° and x° . Find x. [2]<<
>>(d) Classify the triangle (acute / right / obtuse). [2] **[8]**

4. (a) Expand and simplify $(2x - 4)(x + 4)$. [3]<<
>>(b) Factorise $x^2 + 5x + 4$. [2]<<
>>(c) Solve $(x + 4)/2 = 4$. [3]<<
>>(d) Simplify $(x^2 - 16)/(x - 4)$. [2] **[10]**

5. Line L: $y = 3x + 5$. <<
>>(a) Gradient and y-intercept. [2]<<
>>(b) Coordinates where L meets the x-axis. [2]<<
>>(c) Show A(0, 5) lies on L. [2]<<
>>(d) Equation of the line parallel to L through (0, 4). [2] **[8]**

6. A cylindrical tank at Prince Edward has radius 7 cm and height 12 cm. $\pi = 22/7$. <<
>>(a) Volume. [3]<<
>>(b) Curved surface area. [3]<<
>>(c) Total surface area including the base only (open top). [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** A cellphone tower is observed from A on level ground in Masvingo. Angle of elevation of the top is 30° . Distance A to the foot is 24 m. <<
>>(a) Height of the object. Use $\tan 30^\circ = 1/\sqrt{3}$. [4]<<
>>(b) From B, 24 m due East of A, the bearing of the foot is 300° . Sketch. [3]<<
>>(c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**
- 8.** Triangle P(1,1), Q(4,1), R(1,4).<<
>>(a) Reflect in the y-axis. Coordinates of P'. [4]<<
>>(b) Translate PQR by (2 ; -2). Give Q''. [3]<<
>>(c) Matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ maps (x;y). Describe fully and find the image of Q. [5] **[12]**
- 9.** 40 pupils at Prince Edward sat a test (0-50). An ogive is drawn.<<
>>(a) How to read the median from the ogive. [3]<<
>>(b) Median estimated as 34. Meaning? [2]<<
>>(c) How to find the IQR from the graph. [4]<<
>>(d) Why IQR is better than the range here. [3] **[12]**
- 10.** O is the centre. A, B, C on the circumference. AB is a diameter. Angle BAC = 32° .<<
>>(a) Angle ACB. Reason. [3]<<
>>(b) Angle BOC (centre, same arc BC). [3]<<
>>(c) D on the remaining circumference. Angle BDC. Reason. [3]<<
>>(d) State the angle-in-a-semicircle theorem. [3] **[12]**
- 11.** A kombi leaves Masvingo at 08:00 and travels 150 km at constant speed, arriving after 3 hours. It stops 30 minutes, then returns at 80 km/h.<<
>>(a) Outward speed. [3]<<
>>(b) Time for the return. [3]<<
>>(c) Sketch a distance–time graph, labelling axes. [4]<<
>>(d) Average speed for the whole trip including the stop. [2] **[12]**
- 12.** (a) Solve $x^2 - 9x + 18 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Masvingo is 6 m longer than it is wide. If the area is 27 m^2 , form and solve an equation for the width w. [5]<<
>>(c) Sketch $y = (x - 3)(x - 6)$, showing intercepts. [3] **[12]**
- 13.** OA = a = (6 ; 2), OB = b = (4 ; 2). M midpoint of AB.<<
>>(a) Vector AB. [3]<<
>>(b) |AB|. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show AM = MB. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).